



Aerospace Academy Version

Unit 1

Design Process & Aerospace Missions

Unit Overview

Students learn how engineers define mission-based problems, identify stakeholders, criteria, and constraints, research existing solutions, brainstorm concepts, use decision matrices, and present an early aerospace design proposal.

Essential Question

How do aerospace engineers turn a mission need into a testable design solution?

Name: _____ Period: _____ Date: _____

Learning Targets

1. Explain the engineering design process.
2. Identify mission objectives, stakeholders, criteria, and constraints.
3. Write a clear engineering problem statement.
4. Research existing aerospace solutions without copying.
5. Create annotated concept sketches.
6. Use a decision matrix and justify a selected design.
7. Present an early design concept in a design review.

Unit Deliverables

- Mission title and mission objective
- Stakeholder map
- Criteria and constraints chart
- Problem statement
- Research notes
- Brainstorming page
- Three annotated concept sketches
- Trade-off analysis
- Decision matrix
- Final concept justification
- Design review presentation plan
- Individual reflection

Vocabulary

Term	Definition
Engineering Design Process	
Mission Objective	
Stakeholder	
Criteria	
Constraint	
Problem Statement	
Research	
Brainstorming	
Concept Sketch	
Annotation	
Trade-Off	
Decision Matrix	
Prototype	
Iteration	
Design Review	
Payload	

Aerospace Mission Concept Challenge

A hurricane has damaged roads and bridges near a coastal community. Emergency responders need to deliver a small fragile sensor package to a remote location. The sensor package must arrive safely and should be recoverable if possible. Your team will develop an early aerospace-based mission concept and present it in a design review.

Day 1 — What Is Engineering Design?

Objective: I can explain engineering design and describe how engineers solve problems.

Bellwork: What does it mean to design something? Is design only about appearance?

Core Task: Analyze a classroom object: what problem does it solve, who uses it, what makes it successful, and what limits affected it?

Evidence of Work / Notes:

Exit Ticket: How is engineering design different from making something look good?

Day 2 — Aerospace Systems and Missions

Objective: I can describe aerospace systems and how they complete missions.

Bellwork: List five things aerospace engineers design besides airplanes.

Core Task: Break an aerospace system into subsystems and explain what each subsystem does.

Evidence of Work / Notes:

Exit Ticket: What is the difference between a system and a subsystem?

Day 3 — Mission Objective

Objective: I can identify the mission objective for an aerospace design problem.

Bellwork: Why must engineers understand the mission before designing?

Core Task: Draft and revise a mission objective for the hurricane sensor delivery scenario.

Evidence of Work / Notes:

Exit Ticket: How is a mission objective different from a design idea?

Day 4 — Stakeholders, Criteria, and Constraints

Objective: I can identify stakeholders, criteria, and constraints.

Bellwork: Who might be affected by an aerospace design besides the engineer?

Core Task: Create a stakeholder map and criteria/constraints chart for the mission.

Evidence of Work / Notes:

Exit Ticket: Write one criterion and one constraint for the mission.

Day 5 — Problem Statements

Objective: I can write a clear engineering problem statement.

Bellwork: Why is 'build something that works' not a strong problem statement?

Core Task: Use the problem statement builder to write and revise a mission-focused problem statement.

Evidence of Work / Notes:

Exit Ticket: What should a strong problem statement include?

Day 6 — Research Without Copying

Objective: I can research existing solutions and record useful design information.

Bellwork: Why should engineers research before brainstorming?

Core Task: Research at least three related aerospace or emergency response solutions and summarize useful design features.

Evidence of Work / Notes:

Exit Ticket: What is one useful idea you found during research?

Day 7 — Brainstorming Possible Solutions

Objective: I can generate multiple possible solutions before choosing one.

Bellwork: Why should engineers avoid choosing their first idea too quickly?

Core Task: Generate at least eight rough ideas and select three ideas worth developing.

Evidence of Work / Notes:

Exit Ticket: What idea did your team generate that surprised you?

Day 8 — Concept Sketching

Objective: I can create concept sketches that communicate design ideas.

Bellwork: Why are sketches useful before a design is final?

Core Task: Create three concept sketches with labels, payload notes, recovery/safety notes, and concerns.

Evidence of Work / Notes:

Exit Ticket: What annotation made your sketch clearer?

Day 9 — Annotations and Design Intent

Objective: I can improve sketches using annotations that explain design intent.

Bellwork: What might someone misunderstand if a sketch has no labels?

Core Task: Upgrade your strongest sketch with at least six annotations and explain three design choices.

Evidence of Work / Notes:

Exit Ticket: What is the difference between a label and an annotation?

Day 10 — Trade-Offs in Aerospace Design

Objective: I can explain trade-offs in aerospace design decisions.

Bellwork: What matters more: lightweight, strong, low-cost, or reliable? Explain.

Core Task: Compare three concepts by strengths, weaknesses, and biggest trade-offs.

Evidence of Work / Notes:

Exit Ticket: Describe one trade-off your team must consider.

Day 11 — Decision Matrix

Objective: I can use a decision matrix to compare design concepts.

Bellwork: Why should engineers use evidence instead of only voting for a favorite idea?

Core Task: Choose five criteria, score three concepts, and identify the highest scoring concept.

Evidence of Work / Notes:

Exit Ticket: Why is a decision matrix useful?

Day 12 — Final Concept Justification

Objective: I can justify a selected design using evidence.

Bellwork: What makes a justification stronger than 'we liked it'?

Core Task: Write a final concept justification using mission objective, research, sketches, trade-offs, and the decision matrix.

Evidence of Work / Notes:

Exit Ticket: What evidence best supports your final design choice?

Day 13 — Design Review Preparation

Objective: I can prepare a design review presentation.

Bellwork: Why do engineers present unfinished designs during a design review?

Core Task: Plan the design review slides and assign team responsibilities.

Evidence of Work / Notes:

Exit Ticket: Which slide is most important for explaining your team's thinking?

Day 14 — Peer Feedback and Revision

Objective: I can give and use feedback to improve a design review.

Bellwork: What makes feedback useful instead of critical?

Core Task: Practice with another team, record one strength, one question, and one suggestion, then revise.

Evidence of Work / Notes:

Exit Ticket: What change did your team make because of feedback?

Day 15 — Design Review Presentation

Objective: I can present an early aerospace design concept and reflect on design process.

Bellwork: What should a professional presenter do during a design review?

Core Task: Present the design review and complete the individual reflection.

Evidence of Work / Notes:

Exit Ticket: What would your team test first if the project continued?

Mission Concept Portfolio Rubric

Category	Points
Mission objective and problem statement	10
Stakeholders, criteria, and constraints	10
Research notes	10
Brainstorming and concept development	10
Concept sketches and annotations	15
Trade-off analysis	10
Decision matrix	10
Final concept justification	10
Design review presentation	10
Individual reflection	5
Total	100

Unit 1 Student Work Templates

Research Notes Template

Purpose: Research without copying

Directions: Record useful ideas from existing solutions. Focus on features, constraints, materials, failure points, and trade-offs.

Topic/source: _____

Key idea: _____

Useful features: _____

Limits/risks: _____

How this could inspire our design:

Notes:

Research Notes Template

Purpose: Research without copying

Directions: Record useful ideas from existing solutions. Focus on features, constraints, materials, failure points, and trade-offs.

Topic/source: _____

Key idea: _____

Useful features: _____

Limits/risks: _____

How this could inspire our design:

Notes:

Research Notes Template

Purpose: Research without copying

Directions: Record useful ideas from existing solutions. Focus on features, constraints, materials, failure points, and trade-offs.

Topic/source: _____

Key idea: _____

Useful features: _____

Limits/risks: _____

How this could inspire our design:

Notes:

Concept Sketch 1

Purpose: Concept development

Directions: Use quick, clear shapes. Add labels, arrows, and notes that explain function, materials, safety, payload protection, or design intent.

Concept title: _____

Design goal: _____

Key features: _____

Concept Sketch - Payload Delivery Concept 1

Strongest _____

feature: **Biggest concern:** _____

Concept Sketch 2

Purpose: Concept development

Directions: Use quick, clear shapes. Add labels, arrows, and notes that explain function, materials, safety, payload protection, or design intent.

Concept title: _____

Design goal: _____

Key features: _____

Concept Sketch - Payload Delivery Concept 2

Strongest _____

feature: **Biggest concern:** _____

Concept Sketch 3

Purpose: Concept development

Directions: Use quick, clear shapes. Add labels, arrows, and notes that explain function, materials, safety, payload protection, or design intent.

Concept title: _____

Design goal: _____

Key features: _____

Concept Sketch - Payload Delivery Concept 3

Strongest _____

feature: **Biggest concern:** _____

Decision Matrix Template

Purpose: Evidence-based selection

Directions: Choose criteria that connect to the mission. Score each concept honestly using evidence from research, sketches, constraints, and trade-offs.

Criteria	Weight	Concept 1	Concept 2	Concept 3
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Total

Selected concept: _____

Justification:

Design Review Storyboard

Purpose: Presentation planning

Directions: Plan each slide before building the final presentation. Keep each slide focused on one purpose.

Slide	Purpose / Content	Speaker / Notes
1	Mission title and team	
2	Mission objective and problem statement	
3	Stakeholders, criteria, constraints	
4	Research summary	
5	Concept options	
6	Decision matrix	
7	Final concept and justification	
8	Risks, limitations, next steps	

Presentation reflection notes:
